

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [84, 82, 81, 77, 80, 71, 84, 72, 76, 72]

Tap amplitudes (pixels): [163.52675622050356, 165.95180023127196, 173.21951391226105, 168.04761230080004, 165.7015389186232, 164.45364088398895, 156.33937443907084, 167.60966559241146, 152.50245899656832, 152.11837495845134, 156.81836627130127]

✦ Based on the extracted features provided, here is the detailed analysis and scoring according to the MDS-UPDRS Item 3.4 criteria.

1. Analysis of Tapping Speed and Rhythm (Periods)

- **Data:** [84, 82, 81, 77, 80, 71, 84, 72, 76, 72]
- **Observation:** The periods (time between taps) trend downward, moving from ~84 frames at the start to ~72 frames at the end.
- **Interpretation:** A lower period indicates a higher frequency. The subject actually **increased their speed** (accelerated) slightly throughout the test. There is no evidence of slowing.
- **Interruptions:** The variance between frames is consistent. There are no massive spikes in the period data (e.g., a jump to 150+ frames) that would indicate a hesitation, halt, or freezing of movement.
- **Criterion Check:**
 - Hesitations/Interruptions: **None.**
 - Slowing: **None.**

2. Analysis of Amplitude (Range of Motion)

- **Data:** [163.5, 165.9, 173.2, 168.0, 165.7, 164.4, 156.3, 167.6, 152.5, 152.1, 156.8] (Rounded)
- **Observation:**
 - **Start (Taps 1-3):** Strong opening, peaking at Tap 3 (173.2 px).
 - **Middle (Taps 4-8):** Generally stable with a slight dip at Tap 7 (156.3 px) followed by a strong recovery at Tap 8 (167.6 px).
 - **End (Taps 9-11):** The amplitude drops to its lowest levels in the sequence (152.5, 152.1, 156.8 px).
- **Criterion Check:**
 - There is no decrement starting after the 1st tap (rules out Score 3).
 - There is no consistent decrement midway (Tap 8 recovered significantly, ruling out Score 2).
 - There is a visible reduction in amplitude in the final three taps compared to the peak and the start. This aligns with **Score 1 (c): "The amplitude decrements near the end of the 10 taps."**

3. Conclusion

The subject performed more than the required 10 taps (11 total). The speed was excellent and actually improved over time. The rhythm was stable with no interruptions. However, the amplitude data shows a reduction in the range of motion specifically in the final portion of the test (Taps 9, 10, and 11 are the lowest values in the set).

This specific pattern—good performance that exhibits fatigue/decrement only at the very end of the sequence—matches the definition for a **Slight** impairment.

Score: 1

